

David Lau

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EDUCATION

SUNY Stony Brook University

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Computer Networks, Software Security

Stony Brook, NY

Aug. 2022 – May 2024

EXPERIENCE

Software Engineer

BNY Mellon

July 2024 – Present

New York, NY

- Cut P95 latency $2\times$ (2.4s to 1.2s) and trimmed average from 1.2s to 0.6s by optimizing query execution plans, introducing covering indexes, and implementing batched writes in Oracle and Snowflake, sustained 10k+ transactions/hour without SLA regressions.
- Reduced on-call alerts by 30% by making scheduled reporting jobs idempotent and adding simple retry and dead-letter logic; stabilized the nightly email/reporting pipeline and helped maintain a 35% drop in data-quality incidents across Oracle and Snowflake systems.
- Designed and implemented a Data Quality Monitor leveraging Kafka queues and worker services to validate daily Oracle to Snowflake data transfers in parallel by built consistency checks, automated retries, and alerting via email, which reduced an undetected data issues by 25% and improving pipeline reliability across distributed systems.
- Developed backend logic and services to analyze Snowflake warehouse usage patterns, detect underutilized datasets, and trigger storage optimizations automatically. Wrote Python and SQL modules to compress infrequently accessed data and integrated auto-suspend APIs, lowering data-warehouse costs by 20% with no performance regressions.
- Built end to end data lineage pipelines across Snowflake and Oracle with automated backtracking and reconciliation, converting 8,000+ previously untraced rows into fully traced, audit ready lineage and enabling fully automated, hands free scheduled updates going forward

Software Engineering Intern

BNY Mellon

May 2023 – Aug 2023

New York, NY

- Developed an Angular-based financial portal backed by Oracle to aggregate and visualize earnings across four business sectors; adopted by 1,000+ internal users.
- Implemented automated unit and integration tests, increasing test coverage by 30% and reducing critical bugs.
- Optimized API endpoints, reducing average response time by 25% and improving frontend load times and user experience.
- Refactored core backend services, reducing technical debt by 12% and improving code maintainability.
- Redesigned the internal platform as an Angular Progressive Web App (PWA), boosting mobile adoption and accessibility while decreasing load time by 15%.

PROJECTS

Geodudes | *React.js, Redux, Leaflet, Node.js, Express, MongoDB*

- Developed GeoDudes, a comprehensive mapping platform, utilizing MERN stack, and redux for efficient state management.
- Integrated an advanced heatmap feature to visualize geospatial data density and also intelligently cluster data points, displaying the count within each cluster for a concise representation of large datasets across the globe.
- Automated deployment to render using GitHub Actions that initiates comprehensive testing using Cypress, Supertest, and Jest, ensuring automatic deployments upon successful push to the main branch.

Zoom Clone | *React.js, Next.js, Tailwind, Clerk, Vercel*

- Developed and deployed a zoom clone, achieving seamless and reliable video communication during meetings.
- Implemented features such as private/public meetings, meeting scheduling, meeting recording, and screen sharing.
- Integrated user authentication with Clerk, supporting both registered users and guest mode, ensuring secure access and personalized user experiences.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript, HTML/CSS, SQL, Bash / Shell scripting

Tools & Technologies: Git (GitHub, GitLab), Maven, Spring, AppEngine, MongoDB, OracleDB, Snowflake, AWS

Languages: English, Cantonese